7.6.2 Noise pollution

As drone operations expand across commercial, industrial, and public sectors, noise pollution has emerged as one of the most noticeable and growing environmental challenges. While drones are significantly quieter than traditional aircraft, the high-frequency noise generated by their propellers and motors can be intrusive, particularly in urban areas and natural habitats. This persistent background noise affects not only human populations but also wildlife, which relies heavily on acoustic cues for communication, navigation, and survival. Managing and mitigating drone-generated noise has therefore become a crucial objective in sustainable drone development.

The noise produced by drones primarily stems from rotor blade aerodynamics, motor vibration, and airflow turbulence. Multi-rotor drones, such as quadcopters, produce a continuous buzzing sound that intensifies with speed and proximity. In residential zones, this can cause public annoyance, psychological discomfort, and potential sleep disruption, particularly as the frequency of drone deliveries, aerial inspections, and surveillance operations increases. In wildlife environments, drone noise can disrupt animal communication patterns, trigger stress responses, and interfere with predator–prey interactions. Birds, for example, are especially sensitive to drone sounds, often altering flight or nesting behavior even at moderate altitudes.

To address these issues, researchers and manufacturers are developing multiple strategies for noise reduction and acoustic optimization. One key approach involves redesigning propellers to reduce aerodynamic turbulence. The use of serrated or curved blade edges, inspired by owl wing morphology, has shown promise in lowering noise levels without sacrificing lift efficiency. Similarly, variable-pitch propellers and low-RPM